190 T0 Korrekturen

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Contents

Preliminary Note

Status: Fully incorporated — May 2026

All corrections (K1–K3) and refinements (P1–P13) listed in this document have been incorporated into the respective source documents. The affected documents now stand independently without reference to this errata register.
This document is retained as a historical archive. It is **no longer to be used as a binding reference** — the current state of each source document is authoritative.

This document lists corrections and refinements for previously published documents in the FFGFT series. Older documents are *not* modified. Where a contradiction exists between an older and a newer document, the version recorded here is binding.

Each entry contains: affected document and location, the erroneous expression, the correct expression, and a brief justification.

.1 Correction 1: Prefactor in the Higgs derivation of ξ

Affected Document

Doc. 049 (T0 Lagrangian and Higgs Integration), HTML draft version and earlier working versions.

Erroneous Expression

$$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} \approx 1.0 \times 10^{-5} \qquad \text{(old – wrong)}$$

Correct Expression

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4}} \qquad (190.1)$$

with the experimental inputs:

$$\begin{aligned} m_h &= 125.25\,\text{GeV} && \text{(Higgs mass)},\\ v &= 246.22\,\text{GeV} && \text{(vacuum expectation value)},\\ \lambda_h &= \frac{m_h^2}{2v^2} \approx 0.129 && \text{(Higgs self-coupling)}. \end{aligned}$$

Justification

The prefactor $64\pi^4$ originated as a typo in the early HTML visualization. It yields $\xi \approx 10^{-5}$, which deviates by an order of magnitude from the geometrically derived value $\xi = 4/3 \cdot 10^{-4}$ and does not reproduce the natural constants. The correct prefactor $16\pi^3$ yields $\xi \approx 1.33 \times 10^{-4}$, consistent with the value independently determined from the proton-electron mass ratio (Doc. 009).

.2 Correction 2: Tau-Yukawa prefactor r_τ

Affected Document

Doc. 116 (Koide formula as a consequence of T0 theory), table of Yukawa coefficients.

Erroneous Expression

$$r_\tau = \frac{8}{3} \approx 2.667 \quad \Rightarrow \quad m_\tau \approx 1712\,\text{MeV} \quad (-3.6\,\%\ \text{dev.}) \quad \text{(old – wrong)}$$

Correct Expression

$$\boxed{r_\tau = \frac{25}{9} \approx 2.778} \quad \Rightarrow \quad m_\tau \approx 1783\,\text{MeV} \quad (+0.4\,\%\ \text{dev.}) \tag{190.2}$$

The lepton masses are generally given by:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v \tag{190.3}$$

with exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ (step size $\Delta p = 1/3$) and the vacuum expectation value v.

Justification

$r_\tau = 8/3$ was a remnant from an earlier iteration stage. Doc. 158 (Koide-to-g-2 bridge) already uses $r_\tau = 25/9$, which agrees numerically with the tau mass. Furthermore, $25/9 = (5/3)^2$ has a geometric interpretation as the square of the pure sixth in the natural overtone series, consistent with the musical resonance analogy (Doc. 060).

Complete Corrected Prefactor Table

Lepton	Exponent p	Prefactor r	m [MeV]
Electron	$3/2$	$4/3$	0.511
Muon	1	$16/5$	105.7
Tau	$2/3$	$25/9$	1783

.3 Correction 3: Base formula for g-2 of the electron

Affected Document

T0 Explorer (HTML visualization), first version; earlier draft versions of Doc. 018.

Erroneous Expression

$$a_e = \frac{4\pi}{f \cdot k_{\text{geom}}} \qquad \text{(old – wrong)}$$

Correct Expression

$$\boxed{a_e = \frac{S_3}{f \cdot k_{\text{geom}}} = \frac{2\pi^2}{f \cdot k_{\text{geom}}}} \tag{190.4}$$

with $f = 1/\xi \approx 7500$, $k_{\text{geom}} = (2/\sqrt{\varphi}) \cdot \sqrt{2} \approx 2.224$ and $S_3 = 2\pi^2 \approx 19.74$ (surface area of the 4D unit sphere, i.e. the 3D boundary of the torus).

Justification

The earlier prefactor 4π corresponds to the surface area of the 2D sphere in 3D space, not the geometrically correct 3D surface of the 4D torus. $S_3 = 2\pi^2$ is the correct value from torus geometry. The fractal corrections for muon and tau still use 4π:

$$\Delta a_\mu = \frac{4\pi}{f^{5/3}}, \tag{190.5}$$

$$\Delta a_\tau = \frac{4\pi}{f^{4/3}}. \tag{190.6}$$

.4 Refinement 1: Accuracy of the Koide formula

Affected Documents

Doc. 116 (abstract, line 14): $\Delta Q < 0.00003\,\%$
Doc. 158 (abstract): "experimentally confirmed to $0.001\,\%$"

Correct Interpretation

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.7}$$

is experimentally confirmed to better than $0.001\,\%$ (PDG values 2022). The figure $< 0.00003\,\%$ in Doc. 116 refers to the internal consistency of the T0 formulas, not the experimental measurement uncertainty. Both statements are substantively correct but refer to different quantities. In external communications, $0.001\,\%$ is the appropriate figure.

Note

The individual T0 masses agree with experiment to within $\sim 1\,\%$; the high precision of $Q = 2/3$ arises from the specific algebraic structure of the Koide combination, not from individual mass accuracy.

.5 Refinement 2: $\hbar$ from the Higgs field – derivation chain

Context

Planck's constant $\hbar$ is derived differently in various documents. The complete, consistent derivation chain is fixed here.

Binding Derivation Chain

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \longrightarrow L_0 = \xi \cdot \ell_P \longrightarrow \hbar \sim \sqrt{\xi} \cdot \ell_P \cdot c \tag{190.8}$$

The time-mass binding condition provides the physical justification:

$$T(x,t) \cdot m(x,t) = 1 \implies T = \frac{\hbar}{mc^2} \implies E = \hbar\omega. \tag{190.9}$$

$h = 2\pi\hbar$ is therefore *not* a free constant but a geometric consequence of ξ and ℓ_P. The numerical SI value $h \approx 6.626 \times 10^{-34}$ J·s is the calibration of this relation onto the SI base units.

.6 Refinement 3: Two ξ parameters – two distinct spaces

Context

In Documents 173–176 (quantum dots, spin qubits, qubit state spaces, Shor algorithm), two numerically different ξ values appear. Earlier documents did not separate these consistently, which can cause confusion.

Binding Distinction

Parameter	Value	Domain	Meaning
ξ_0 (geometric)	$\frac{4}{30000} \approx 1.33 \times 10^{-4}$	3D geometric space	Length hierarchy of the torus; valid in physical position space
ξ_{Higgs} (algebraic)	$\approx 1.04 \times 10^{-5}$	Number / quantum state space	Algebraic field corrections; Higgs sector; decoherence rates

Justification

$\xi_0 = 4/30000$ follows from the torus geometry of physical space (geometric derivation, independent of SI measurements). ξ_{Higgs} follows from the algebraic Higgs field structure and appears in coupling formulas between quantum states (e.g. Bell correlations, T_2 hierarchy).

The two ξ values are not a contradiction; they describe complementary aspects: geometry of space (ξ_0) vs. algebra of state space (ξ_{Higgs}). External communications must always specify which parameter is meant.

.7 Refinement 4: L_0 and the Schwarzschild interpretation

Affected Documents

Earlier versions of Doc. 180–182 (Lagrangian derivation, torus geometry, maximum cosmological scale).

Erroneous Approach

Earlier versions interpreted $L_0 = \xi_0 \cdot \ell_P$ as the Schwarzschild radius of a fundamental minimal object. This interpretation was explicitly discarded in Doc. 182 (2026 revision).

Correct Interpretation

L_0 is exclusively the **geometric fundamental length** of the FFGFT torus:

$$L_0 \; = \; \xi_0 \cdot \ell_P \; \approx \; 2.15 \times 10^{-39}\,\text{m}. \tag{190.10}$$

It defines the lowest scale stage of the ξ hierarchy and has *no* Schwarzschild interpretation. The cosmological Hubble radius R_H is the system-specific upper limit of the cosmological sector – not a universal maximum scale. Each scale has its own system-specific upper limit.

Note: Schwarzschild as Consistency Check

The Schwarzschild connection remains valid as an **internal consistency check**: an object of mass $M_0 = \xi_0 \cdot m_P/2$ would have a Schwarzschild radius exactly equal to L_0 – with no free parameter.

If M_0 matches a scale stage of the ξ hierarchy, that is a non-trivial self-consistency test. The connection is *a probe, not a derivation*.

.8 Refinement 5: Scope and reach of the mass formulas

Context

Several FFGFT documents describe mass derivations for leptons and other particles. The scope of these derivations must be communicated clearly.

Binding Formulation

The FFGFT mass formulas (e.g. $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$) yield values that agree with experiment to $\sim 1\,\%$. This is **not a precision calculation** but a structural proof:

- The mass formulas *follow geometrically* from ξ and v – they are not fitted.
- The remaining $\sim 1\,\%$ deviations reflect measurement uncertainties and approximations, not errors in the structure.
- Numerical accuracy beyond the Standard Model is *not* claimed.

To be avoided in external texts: "FFGFT calculates the masses exactly." The correct statement is: "FFGFT *derives* the mass structure from a single parameter ξ and reproduces the orders of magnitude to $\sim 1\,\%$."

.9 Refinement 6: Koide formula only with original values

Binding Rule

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.11}$$

must be evaluated only with experimentally measured masses (PDG values).

Not permitted: substituting the symbolic FFGFT terms $r_\ell \cdot \xi^{p_\ell} \cdot v$ directly into the formula. The $\sqrt{m_i}$ operation generates cross-terms with mixed ξ powers that are foreign to the torus structure space. Any such substitution yields $Q \approx 0.6677 \neq 2/3$ and is not a contradiction to the theory but a category error.

Permitted: evaluating FFGFT terms numerically (e.g. $m_\tau = 1783$ MeV) and inserting that result into the Koide formula like a measured value.

Mathematical background: The r_ℓ terms are geometric invariants of their torus modes. The set of admissible pairs (r_i, p_i) is not closed under multiplication: $r_e \cdot r_\mu = 64/15$ does not belong to any torus mode. This is the Non-Closure Theorem (Doc. 193). The general analysis of composition limits is found in Doc. 192.

.10 Refinement 7: Terminology "renormalization group" and scale structure

Affected Documents

- Doc. 005 (T0 tm-extension): "RG flow", "renormalization group factor", "renormalization group invariance", subsection "renormalization group treatment".
- Doc. 008 (ξ and e): subsection "modified renormalization group equations".
- Doc. 019 (T0 Lagrangian, earlier version): subsection "renormalization group equations".
- Doc. 086 (document overview): "renormalization group as flow in parameter space".
- Doc. 189 (torus derivation): "cumulative renormalization group running".
- Doc. 202 (FFGFT comprehensive field theory), §18: section title "The renormalization group".

Previous Formulation

In the documents listed, the ξ_0-modified scale running of the coupling

$$\frac{d\alpha}{d\ln\mu} = \beta_0\,\alpha^2\left(1 + \xi_0\,\ln\frac{\mu}{E_0}\right) \qquad \text{(old – imprecise)}$$

and the corresponding scale dependence are designated as the *renormalization group*, sometimes adjacent to statements explicitly characterizing ξ_0 as a topologically fixed geometric constant.

Refined Formulation

In FFGFT, the scale dependence of the coupling is *not* a result of a renormalization procedure but a direct consequence of the recursion $\mathscr{R}$ (Doc. 203) on the fractal torus geometry. The correct designation is therefore:

$$\boxed{\begin{aligned}&\text{Scale structure from recursion}\\ &\equiv\ \text{recursive } \alpha \text{ running via } \mathscr{R}\end{aligned}} \qquad (190.12)$$

Specifically:

- The term $\xi_0\,\ln(\mu/E_0)$ arises from the iterated application of the recursion $\mathscr{R}$ to the mode structure, not from a counter-term subtraction.
- ξ_0 is *not* a fixed point of a β function but a topologically fixed constant ($\xi_0 = 4/30000$, Doc. 180).
- Stability, scale flow, and UV finiteness are three aspects of *the same* recursive structure – not separate mechanisms.

Language Convention

Avoid	**Use**
Renormalization group (implies QFT procedure)	Scale structure from recursion
RG flow / RG running	Recursive α running; $\mathscr{R}$-induced scale correction
Renormalization group equations	Scale equations from $\mathscr{R}$
Renormalization group invariance	Recursive scale invariance
Renormalization group fixed point	Recursion fixed point: $\mathscr{R}(\Phi_*) = \Phi_*$

Justification

The designation “renormalization group” imports a tool of standard quantum field theory based on a fundamentally different principle: there UV divergences are absorbed by a procedure, and parameters *run* under a formal scale transformation. In FFGFT neither the divergence problem (cf. Doc. 159, “without artificial renormalization”) nor a free scale parameter to be absorbed exists. The scale dependence is a *structural property* of the recursion, not a subsequent adjustment step.

The FFGFT-native view is already correctly formulated in Doc. 159 (“renormalization as symptom, not as solution”) and Doc. 189 (“geometrically determined, not by renormalization”); this clarification serves to unify the terminology across the older Lagrangian and ξ-derivation series (Doc. 005, 008, 019) and across Doc. 202.

The underlying mechanism is developed in Doc. 203 (recursion operator $\mathscr{R}$) and Doc. 204 (fractal boundary condition).

.11 Refinement 8: Terminology “fractal renormalization”

Affected Documents

Doc. 005, 008, 012, 014 (with 19 occurrences the most frequent source), 041, 060, 133, 141, 159, 181.

Previous Formulation

“Fractal renormalization” is used as an FFGFT-internal term for the geometric correction factors in the conversion between natural and SI units, and for scale-dependent corrections on the fractal torus.

Refined Formulation

Since the word “renormalization” – even in compound form – carries the QFT connotation of a subtraction or adjustment procedure, the term is replaced by context-specific alternatives:

Context	Binding designation
Unit conversion natural ↔ SI (esp. Doc. 014)	Geometric scale adjustment
Scale-dependent corrections on the fractal torus (Doc. 133, 159, 181)	Fractal correction (matches title of Doc. 133)
General scale transformation by recursion	$\mathscr{R}$-induced scale correction

Justification

The proprietary term “fractal renormalization” was meant to distinguish the procedure clearly from QFT renormalization, but it adopts the central word and unavoidably with it the connotation of a *correction procedure*. In FFGFT this is not a procedure but the direct numerical manifestation of the fractal geometry. The proposed alternatives avoid this misunderstanding without requiring new vocabulary – both phrases are already established in the corpus (Doc. 133 “fractal correction”; “geometric scale adjustment” in the same semantic field as Doc. 014).

.12 Refinement 9: Column “Symmetry” and the term “color charge” in the fermion tables

Affected Documents

- Doc. 006 (T0 particle masses), “universal fermion table”, column “Symmetry”.
- Doc. 046 (Particle Masses, English parallel version), same table, column “Color”.
- Doc. 042 (ξ parameter and particles), table “spatial field patterns”, entry “Quarks: multi-node bound clusters, confined, color charge”.
- Doc. 005 (T0 tm-extension), Method 2 “extended fractal method with QCD integration”: use of $N_c = 3$, α_s, Λ_{QCD} as external inputs.

Previous Formulation

In the fermion tables of Doc. 006/046 each row carries a column "Symmetry" with the following entries:

Particle	Entry under "Symmetry"
Electron, Muon, Tau	Lepton number
ν_e, ν_μ, ν_τ	Double-ξ
Up, Charm, Strange, Top, Bottom	Color
Down	Color + Isospin

This mixture combines four different concepts in one column: an FFGFT scaling class ("Double-ξ"), an SM conserved quantity ("lepton number"), an SM gauge symmetry ("color"), and an SM quantum number ("isospin"). In addition, the entry "color charge" in Doc. 042 and $N_c = 3$ in Doc. 005 suggest that the SM color symmetry $SU(3)_C$ is an independent component of FFGFT.

Refined Formulation

In FFGFT neither the gauge group $SU(3)_C$ nor a color quantum number exists. Quark masses follow, just like lepton masses, exclusively from the Yukawa resonance formula

$$m_f = r_f \cdot \xi^{p_f} \cdot v \qquad (190.13)$$

with r_f a rational prefactor and p_f a rational exponent (step size $\Delta p = 1/3$). This is verified numerically by the script `calc-resonanz-leiter.py`: all six quarks hit their PDG masses to ~ 0.3–$3.4\,\%$ *without* use of N_c, α_s, Λ_{QCD}, or any color quantum number.

Corrected column structure of the fermion table

The old column "Symmetry" is replaced by two clearly separated columns:

Particle	FFGFT scaling class	SM correspondence (informational)
Electron, Muon, Tau	Single-ξ	Lepton number
ν_e, ν_μ, ν_τ	Double-ξ	Lepton number, lepton mixing
Up, Charm, Top	Cluster-ξ	Up-type, $T_3 = +1/2$
Down, Strange, Bottom	Cluster-ξ	Down-type, $T_3 = -1/2$

Meaning of the classes:

- *Single-ξ*: $m = r\xi^p v$ with $p \in \{2/3, 1, 3/2\}$.
- *Double-ξ*: $m_\nu = r_\nu \xi^2 m_e$ (additional ξ suppression relative to the charged leptons).
- *Cluster-ξ*: $m = r\xi^p v$ with the same exponents $p \in \{-1/3, 1/2, 2/3, 1, 3/2\}$ as for the leptons, but with multi-node binding in the sense of Doc. 049 (cubic self-interaction $\lambda_s(\delta m_q)^3$).

Language convention for quarks

Avoid	Use
"color charge" (implies SU(3)$_C$ quantum number)	Multi-node binding
"three colors red/green/blue" (SM gauge index)	Triple-node cluster (the triplet is geometric, not an $SU(3)$ index)
"$N_c = 3$ colors"	$N_{\text{cluster}} = 3$ (geometric node count in the baryon)
"color confinement"	Cluster binding (cubic self-interaction)

Status of Doc. 005, Method 2

Doc. 005 contains two calculation methods:

- *Method 1: Direct geometric method.* Yukawa form according to Eq. (190.13); parameter-free; accuracy $\sim 1\,\%$ for all fermions. **This is the FFGFT method.**

- *Method 2: Extended fractal method with QCD integration.* Uses N_c, α_s, Λ_{QCD}, lattice-QCD calibration and ML. **This is a phenomenological bridge construction** for connecting to the lattice-QCD literature, not part of the parameter-free FFGFT.

In external communication Method 1 is to be cited. Method 2 may be used if explicitly labelled as "bridge to lattice QCD" or "ML-calibrated extension".

Justification

The designations "color", "color charge", "N_c" import a concept from standard quantum field theory ($SU(3)_C$ as gauge symmetry, three colors, eight gluons, color confinement as a non-perturbative property). Doc. 049 explicitly states that FFGFT contains none of these elements: "what we call ‚color' becomes high-energy node binding patterns" (Doc. 049, §3.7), with a single cubic interaction term $\lambda_s(\delta m_q)^3$ instead of eight gluon fields.

The script validation (`calc-resonanz-leiter.py`, `calc_De.py`) shows: all quark masses follow from the same Yukawa formula as the leptons. The column "Symmetry" of the fermion table, however, suggests that $SU(3)_C$ is a distinguishing property of quarks within FFGFT – which is not the case.

The corrected column structure separates what is FFGFT-proper (*scaling class*) from informational SM correspondence (*SM correspondence*), in parallel with the separation of "recursive α running" and "renormalization group" in Refinement 7.

.13 Refinement 10: Logical status of $K = 245$ and the factor 1.314 in Doc. 009

Affected Document

Doc. 009 (Origin of ξ), Section "Why $K = 245$ is fundamental", subsection "Geometric meaning".

Misleading Presentation

Doc. 009 lists the following derivation chain for $K = 245$:

- $\varphi^{12} = 321.996$
- Correction by fractal structure: $(1-\xi)^2 \approx 0.999733$
- Result: $321.996 \times 0.999733 \approx 321.87$
- "Scaling to mass range": $321.87/1.314 \approx 245$

The factor 1.314 appears here without derivation or independent physical justification.

Correct Logical Order

$K = 245$ is not derived from φ^{12} and 1.314. The correct logical order is the reverse:

$$K = \frac{R_{pe}}{(4/3)^{\kappa}} = \frac{1836.153}{(4/3)^7} \approx 245.097 \tag{190.14}$$

K is fixed by the observed proton-electron mass ratio R_{pe} and the exponent $\kappa = 7$. It is an *empirical anchor*, not a first-principles result. The factor 1.314 is implicitly defined as $\varphi^{12}/245 \approx 1.314$ and does not constitute an independent derivation of K.

What remains non-trivial

The exponent $\kappa = 7$ is the *unique integer* for which the complete e-p-μ triangle closes simultaneously:

κ	$K \times (4/3)^{\kappa}$	Agreement with R_{pe}
6	1376.6	$-25\,\%$
7	1835.4	$0.04\,\%$
8	2447.2	$+33\,\%$

This uniqueness is a genuine constraint, not a tautology. $K = 245$ then follows from $\kappa = 7$ and the empirical R_{pe}.

Corrected claim for external communication

- **Avoid:** "$K = 245$ is derived from first principles; no free parameters."
- **Use:** "One empirical anchor: $K = 245$, fixed by R_{pe} and $\kappa = 7$. From this anchor, five mass ratios follow with $< 0.04\,\%$ accuracy. The φ^{12} observation is a numerological consistency note, not a derivation."

.14 Refinement 11: T_{CMB} correction factor and K_{frak} disambiguation

Affected Documents

Doc. 025 (Cosmology), Doc. 003 (Foundations), Doc. 133 (Fractal Correction Derivation).

Issue A: K_{frak} does not close the T_{CMB} gap

Doc. 025 derives:

$$T_{\text{CMB}} = \frac{16}{9}\xi = \frac{4}{16875}\,\text{eV} \approx 2.7507\,\text{K} \tag{190.15}$$

The observed value is $T_{\text{CMB}} = 2.72548\,\text{K}$ (Planck 2018). The gap is $\approx 0.925\,\%$.

$K_{\text{frak}} = 1 - 100\xi = 74/75$ (Doc. 133) does *not* close this gap. Numerically:

$$\frac{16}{9}\xi \times K_{\text{frak}} = 2.714\,\text{K} \quad (0.42\,\%\ \text{below target, worse}) \tag{190.16}$$

The required correction factor is 0.99083, which satisfies:

$$\boxed{T_{\text{CMB}} = \frac{16}{9}\xi\left(1 - \frac{275}{4}\xi\right) = 2.725486\,\text{K}} \tag{190.17}$$

Agreement with observation: $\Delta = 0.0002\,\%$.

The coefficient $275/4 = 68.75$ is close to the tetrahedral number $68 = 72 - 4$ (Doc. 003, Section 0.5) but its derivation as a second-order correction is not yet formally established. **This is an open point:** a geometric derivation of $275/4$ is required. The formula above is a numerically verified result pending full justification.

Issue B: Two inconsistent K_{frak} definitions

Doc. 003 gives:

$$K_{\text{frak}}^{(003)} = 1 - \frac{D_{\text{frak}} - 2}{68} = 1 - \frac{0.94}{68} \approx 0.98530 \tag{190.18}$$

where $D_{\text{frak}} = 2.94$ is stated without derivation from ξ.

Doc. 133 gives:

$$K_{\text{frak}}^{(133)} = 1 - 100\xi = \tfrac{74}{75} \approx 0.98667 \tag{190.19}$$

derived from the cumulative RG run with $D_f = 3 - \xi$.

These differ by 1.37×10^{-3} and cannot both be correct. The **primary definition** is Doc. 133: $K_{\text{frak}} = 1 - 100\xi = 74/75$. The value $D_{\text{frak}} = 2.94$ in Doc. 003 is an independent approximation not derived from ξ; it uses a different (tetrahedral) model. Doc. 003 should be understood as a heuristic motivation, not as the defining formula.

Corrected claim for T_{CMB}

- **Avoid:** "K_{frak} closes the T_{CMB} gap."
- **Use:** "T_{CMB} requires a second-order correction $(1 - 275/4 \cdot \xi)$, giving $0.0002\,\%$ agreement. The geometric origin of $275/4$ is an open derivation."

Addendum: ΔCHSH $\approx 10^{-5}$ in Doc. 230

Doc. 230 states ΔCHSH $\approx 10^{-5}$ as a geometrically motivated prediction. Doc. 022 gives the formula

$$\text{CHSH}(N) = 2\sqrt{2} \cdot \exp\left(-\xi \frac{\ln N}{D_f}\right) \tag{190.20}$$

where N is the number of measurement runs. This formula gives ΔCHSH $\approx 5 \times 10^{-4}$ at $N = 73$ (not 10^{-5}). Moreover, ΔCHSH grows monotonically with N; at $N = 2$ (the minimum) one already obtains ΔCHSH $\approx 8.7 \times 10^{-5}$. The value 10^{-5} from Doc. 230 is not reachable at any finite N under the Doc. 022 formula.

Key distinction (raised by Onur Teker, May 2026): CHSH$(N) \to 0$ (correlation vanishing) requires $N \to \infty$ and is a different condition from ΔCHSH $= 10^{-5}$ (distance from the Tsirelson bound). Both conditions confirm that the Doc. 022 formula cannot reproduce the Doc. 230 claim. The ΔCHSH $\approx 10^{-5}$ prediction requires a separate derivation or suppression mechanism not yet documented. **Open point.**

.15 Refinement 12: Two distinct ΔCHSH observables in Doc. 022 and Doc. 230

Affected Documents

Doc. 022 (QFT-ML Addendum), Doc. 230 (Hilbert Space Translation).

The apparent contradiction

Doc. 022 gives ΔCHSH $\approx 5 \times 10^{-4}$ at $N = 73$ via the formula CHSH$(N) = 2\sqrt{2} \cdot \exp\left(-\xi \ln N/D_f\right)$. Doc. 230 states ΔCHSH $\approx 10^{-5}$ without derivation. These differ by a factor of ~ 50.

Proposed resolution: cumulative vs. elementary deviation

The two values refer to different observables:

- **Doc. 022** describes the *cumulative* deviation accumulated over N measurement runs. It grows monotonically with N: ΔCHSH$(N) \sim \xi \ln(N)/D_f \times 2\sqrt{2}$. This is the experimentally relevant quantity for a Bell test with N runs.
- **Doc. 230** plausibly refers to the *elementary* deviation per single measurement — the phase cost of one θ_4 projection cycle. The natural candidate is

$$\Delta\text{CHSH}_{\text{elem}} = \frac{\xi}{2\pi} \approx 2.1 \times 10^{-5} \approx 10^{-5} \tag{190.21}$$

  This is the Landauer cost per measurement expressed as a CHSH deviation: the θ_4 phase runs through a full 2π cycle per measurement, and each cycle costs ξ in geometric units.

The ratio between cumulative and elementary deviation at N runs is:

$$\frac{\Delta\text{CHSH}(N)}{\Delta\text{CHSH}_{\text{elem}}} \approx N \cdot \frac{2\pi}{D_f} \tag{190.22}$$

At $N = 73$ this gives a factor of ≈ 153, consistent with the observed factor of ~ 50–100 between the two values.

Status

This resolution is a **working hypothesis**, not a proven derivation. The identification $\Delta\text{CHSH}_{\text{elem}} = \xi/(2\pi)$ requires formal justification from the T⁴ projection geometry. If confirmed, there is no contradiction between Doc. 022 and Doc. 230 — they describe the same physical effect at different integration scales.

Experimental consequence: the cumulative deviation $\Delta\text{CHSH}(N) \approx 5 \times 10^{-4}$ at $N = 73$ is within the reach of next-generation loophole-free Bell tests (current resolution $\sim 10^{-2}$; required improvement: factor ~ 20). The elementary deviation $\xi/(2\pi) \approx 2 \times 10^{-5}$ would require single-shot precision beyond current technology.

.16 Refinement 13: Bridge formula between ξ_{FFGFT} and ξ_{UIFT}

Context

In the IPI discussion (May 2026), Onur Teker (UIFT) writes:

> $\xi = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$, derived from three independent measurements (T_CMB, k_B, m_e). This is not a hyperparameter — it is a prediction.

This appears to identify ξ_{UIFT} with $\xi_{\text{FFGFT}} = 4/30000$. Numerically they are very different. This refinement documents the exact relation.

The key: FFGFT uses eV as temperature unit

In FFGFT, the natural unit system sets $\hbar = c = k_B = 1$. Temperatures are expressed in eV (not in Kelvin):

$$T_{\text{CMB}}[\text{eV}] = k_B \cdot T_{\text{CMB}}[\text{K}] = k_B \cdot 2.72548\,\text{K} = 2.3486 \times 10^{-4}\,\text{eV} \quad (190.23)$$

The FFGFT formula (Doc. 025) then reads — to first order:

$$T_{\text{CMB}}[\text{eV}] = \frac{16}{9}\xi_{\text{FFGFT}} = 2.3704 \times 10^{-4}\,\text{eV} \qquad (\Delta = 0.93\,\%) \quad (190.24)$$

and to second order (Doc. 190 R11):

$$T_{\text{CMB}}[\text{eV}] = \frac{16}{9}\xi_{\text{FFGFT}}\left(1 - \frac{275}{4}\xi_{\text{FFGFT}}\right) = 2.3486{\times}10^{-4}\,\text{eV} \qquad (\Delta = 0.0002\,\%) \quad (190.25)$$

Bridge formula

Substituting (190.24) into Onur's expression (first order):

$$\xi_{\text{UIFT}} = \frac{k_B T_{\text{CMB}}[\text{K}]\ \ln 2}{m_e c^2} = \frac{T_{\text{CMB}}[\text{eV}] \cdot \ln 2}{m_e c^2/\text{eV}}$$
$$= \frac{\frac{16}{9}\,\xi_{\text{FFGFT}} \cdot \ln 2}{m_e c^2/\text{eV}} \tag{190.26}$$

Rearranging:

$$\boxed{\frac{\xi_{\text{FFGFT}}}{\xi_{\text{UIFT}}} = \frac{m_e c^2/\text{eV}}{\frac{16}{9} \cdot \ln 2} = \frac{510\,999\,\text{eV}}{1.2323\,\text{eV}} \approx 414\,684} \tag{190.27}$$

SI conversion table (reconstructed from Doc. 013, archived version)

The following table was documented in older versions of Doc. 013 (SI conversion tables for the FFGFT natural unit system, subsequently condensed). It is reproduced here because it is essential for the bridge formula derivation.

Quantity	Natural units	SI / eV
Temperature unit $[T]$	$= [E]$	1 eV $= k_B^{-1} \cdot 1\,\text{eV} = 11{,}604\,\text{K}$
T_{CMB} [nat.]	$(16/9)\cdot\xi = 2.370\times10^{-4}$	2.370×10^{-4} eV
T_{CMB} [K] (Planck 2018)	—	2.72548 K
T_{CMB} [eV] $= k_B \cdot T$[K]	—	2.3486×10^{-4} eV
$m_e c^2$	$= m_e$ (nat.)	$510{,}999$ eV $=$ 0.511 MeV
k_B	$= 1$ (nat.)	8.617×10^{-5} eV/K
$\hbar$	$= 1$ (nat.)	6.582×10^{-16} eV·s
c	$= 1$ (nat.)	2.998×10^{8} m/s

Numerical verification

Quantity	Value	Unit
$\xi_{\text{FFGFT}} = 4/30000$	1.3333×10^{-4}	dimensionless
$\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$	3.1858×10^{-10}	dimensionless
Ratio $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}}$	$418\,521$	—
Scale factor $m_e c^2/(\frac{16}{9} \ln 2)$ [eV]	$414\,684$	—
T_{CMB} [K] (Planck 2018)	2.72548	K
T_{CMB} [eV] $= k_B \cdot T_{\text{CMB}}$[K]	2.3486×10^{-4}	eV
$(16/9) \cdot \xi$ [eV] (FFGFT 1st order)	2.3704×10^{-4}	eV
$(16/9) \cdot \xi(1 - 275/4 \cdot \xi)$ [eV] (R11)	2.3486×10^{-4}	eV
$m_e c^2$	$510\,999$	eV

The residual difference of $0.93\,\%$ between the ratio ($418\,521$) and the scale factor ($414\,684$) is exactly the first-order T_CMB error — confirming that the bridge formula is correct.

Physical interpretation

ξ_{FFGFT} and ξ_{UIFT} are **not the same quantity**. They are related by a pure SI conversion factor: the ratio of the relativistic electron energy ($m_e c^2 = 511$ keV) to the eV scale of the FFGFT temperature formula, modulated by $(16/9) \cdot \ln 2$.

The SI conversion table above was present in earlier versions of Doc. 013 (archived March 2026, Git commit `f2be9c8`) and was subsequently condensed. The explicit statement $T_{\text{CMB}}^{\text{nat}} = k_B \times 2.725$ K $= 2.35 \times 10^{-4}$ eV appears verbatim in that version. The reconstruction here restores the connection between the natural-unit formula and the SI value that was implicitly assumed but no longer shown.

The Vopson-Teker experiment measures ξ_{UIFT} directly. If confirmed, it does not by itself confirm $\xi_{\text{FFGFT}} = 4/30000$ — it confirms the thermodynamic ground state at T_{CMB}. The geometric derivation of ξ_{FFGFT} from $\kappa = 7$ (Doc. 009) is a separate and independent constraint.

Updated Summary

No.	Doc.	Erroneous / unclear	Correct / refined
C1	049 (HTML)	$\xi = \lambda_h^2 v^2/(64\pi^4 m_h^2)$	$\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$
C2	116 (table)	$r_\tau = 8/3$	$r_\tau = 25/9$
C3	018 (Explorer)	$a_e = 4\pi/(f \cdot k)$	$a_e = 2\pi^2/(f \cdot k)$
R1	116 / 158	$\Delta Q < 0.00003\,\%$ vs. $0.001\,\%$	Both correct; $0.001\,\%$ for external use
R2	several	$\hbar$ as a postulate	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$
R3	173–176	A single ξ value	Two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)
R4	180–182	L_0 as Schwarzschild radius	L_0 = geom. fundamental length
R5	several	"FFGFT calculates masses exactly"	Mass structure from ξ; $\sim 1\,\%$ agreement
R6	116 / 158	Koide with FFGFT symbolic terms	Koide only with numerical values (non-closure, Doc. 193)
R7	005, 008, 019, 086, 189, 202	"Renormalization group" as FFGFT tool	Scale structure from recursion $\mathscr{R}$
R8	005, 008, 012, 014, 041, 060, 133, 141, 159, 181	"Fractal renormalization" as proprietary term	Geometric scale adjustment / fractal correction
R9	005, 006, 042, 046	Column "Symmetry" with "Color"; $N_c = 3$ as external SM input	Columns "Scaling class" + "SM correspondence"; quarks via $m = r\xi^p v$
R10	009	$K = 245$ derived from $\varphi^{12}/1.314$; "no free parameters"	$K = R_{pe}/(4/3)^7$: one empirical anchor; factor 1.314 implicitly defined, not derived
R11	025, 003, 133	K_{frak} closes T_{CMB} gap; one definition of K_{frak}	Correct: $(1 - 275/4 \cdot \xi)$, $\Delta = 0.0002\,\%$; Doc. 003 and Doc. 133 use different models